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ORIGINAL ARTICLE

Enhancing physical-mechanical properties and flame retardancy of fir with lignin-based aqueous modifier

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Abstract

A lignin-based wood modification strategy was proposed by impregnating a composite aqueous modifier consisting of allylated lignin (AL), unsaturated phytic acid amine (TEPA-GMA-PA), and N-hydroxymethyl acrylamide (NMA). The system was designed to enhance the dimensional stability, mechanical properties and flame retardancy of fast-growing fir. AL was synthesized through etherification of industrial alkali lignin with allyl glycidyl ether, endowing the lignin with reactive unsaturated sites for crosslinking. Meanwhile, TEPA-GMA-PA was prepared via a ring-opening addition reaction between glycidyl methacrylate and tetraethylenepentamine, followed by neutralization with phytic acid. The successful synthesis of AL and TEPA-GMA-PA was demonstrated by Fourier Transform Infrared Spectroscopy and Nuclear magnetic resonance spectroscopy. The resulting compound modifier was blended with NMA to form an aqueous polymerizable system. Leaching test demonstrated that the modifier underwent in situ polymerization within the wood structure to form water-insoluble macromolecules, reducing the leaching rate to 34.9%. Modified wood exhibited significant mechanical enhancement with 30.8% and 19.6% increases in modulus of rupture and modulus of elasticity, respectively, while maintaining impact strength to natural wood. Flame retardancy improvements were confirmed by cone calorimetry showing 35.6% reduction in total heat release and 7.2% decrease in total smoke production. Furthermore, the limiting oxygen index increased from 22.6 to 34.7%. The proposed strategy provided a new approach for improving of high-value utilization of wood with lignin-based modifier.

1 Introduction

With the growing demand for sustainable development and environmentally friendly materials, fast-growing wood has emerged as a key renewable resource in fields such as construction and furniture manufacturing due to its easy of processing and abundant availability. However, fast-growing wood (such as fir, radiata pine, poplar and basswood) generally suffers from poor dimensional stability, low mechanical strength and susceptibility to fire, which restrict its high-value application. Wood modification using organic monomers is an effective approach to improve these deficiencies (Chen et al. 2020; Gérardin 2016). The method involves impregnating wood with monomers followed by in-situ free radical polymerization, typically initiated by thermal catalysis, initiators, or irradiation, which can significantly enhance the dimensional stability and durability of the wood (Gérardin 2016). Commonly used organic monomers include styrene (Che et al. 2019), methyl methacrylate (Ghorbani et al. 2017) and vinyl acetate (Ghorbani et al. 2019), etc. These monomers are generally non-polar and have limited compatibility with wood. At the same time, these monomers are toxic and volatile, the modification process needs to be wrapped in aluminum foil to reduce the volatility of monomers, and long time and high temperature polymerization reaction may also lead to degradation of wood components, thereby leading a reduction in wood toughness. In contrast, water-soluble monomers, which contain hydroxyl groups, are more compatible with wood and facilitate penetration into the cell wall, thereby improving modification efficiency. It has been shown that the use of N-isopropylacrylamide/2-hydroxyethyl methacrylate

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can form improve wood dimensional stability by in situ polymerization of water-soluble vinyl monomers. Modified wood has achieved anti-swelling efficiencies (ASE) of 50% (Huang et al. 2019).

Meanwhile, lignin is the most abundant natural aromatic polymer in nature and the main component of plant skeleton (Karthäuser et al. 2021). Alkali lignin is the main by-product of plant fiber used for pulp and paper in industry, containing a large number of phenylpropane units, exhibiting high carbon content, thermal stability, and char-forming ability, which can be used as a biomass-based flame retardant (Yang et al. 2020). The application of lignin as a biomass-based wood modifier is not only in line with the green development needs of the wood industry, but also provides a new idea of the high-value utilization of lignin, which can alleviate the pressure of the traditional wood modification technology relying on fossil resources. Unmodified lignin mainly acts as a char source, offering limited flame retardancy. Combining lignin with flame-retardant additives can enhance its efficiency. Phytic acid (PA) is an organophosphorus compound extracted from grains, seeds and other natural plants, which has the advantages of eco-friendliness and no-toxicity, etc. (Mokhena et al. 2022; Lee et al. 2025). The six phosphate groups in its structure confer strong dehydration and char-forming capability. Upon combustion, PA produced phosphoric or polyphosphoric acid, which catalyzes substrate dehydration and charring. Furthermore, neutralizing PA with amines reduces its acidity, mitigating the negative impact on the mechanical properties of modified wood (Yu et al. 2024).

In this study, a modification strategy was proposed by introducing unsaturated bonds into lignin and PA. Allylated lignin (AL) was synthesized via etherification of industrial alkali lignin with allyl glycidyl ether, incorporating allyl groups into lignin. Simultaneously, an unsaturated PA amine (TEPA-GMA-PA) was synthesized by ring-opening addition of glycidyl methacrylate and tetraethylenepentamine, followed by reaction with PA to enhance its crosslinking capability. These components were then blended with NMA to form a lignin-based composite modifier. The effects of the modifier on the dimensional stability, mechanical performance, and flame retardancy of the modified wood were systematically investigated (Fig. 1), providing insights into property enhancement and flame-retardant mechanisms.

2 Materials and methods

2.1 Materials

Fir was purchased from Fujian Province, China. The density of the fir sapwood was approximately 0.32 g cm^{-3}. Industrial alkali lignin was purchased from Feihuang Chemical Company, Limited (Xinyi, China). N-(Hydroxymethyl) acrylamide (NMA), Tetraethylenepentamine (TEPA), Glycidyl methacrylate (GMA), Allyl glycidyl ether (AGE), Ammonium persulfate (APS) and Phytic acid (PA, 70 wt%) were purchased from Aladdin (Shanghai, China). Anhydrous ethanol and Hydrochloric acid (HCl) were purchased from Hushi (Shanghai, China). Sodium hydroxide (NaOH)

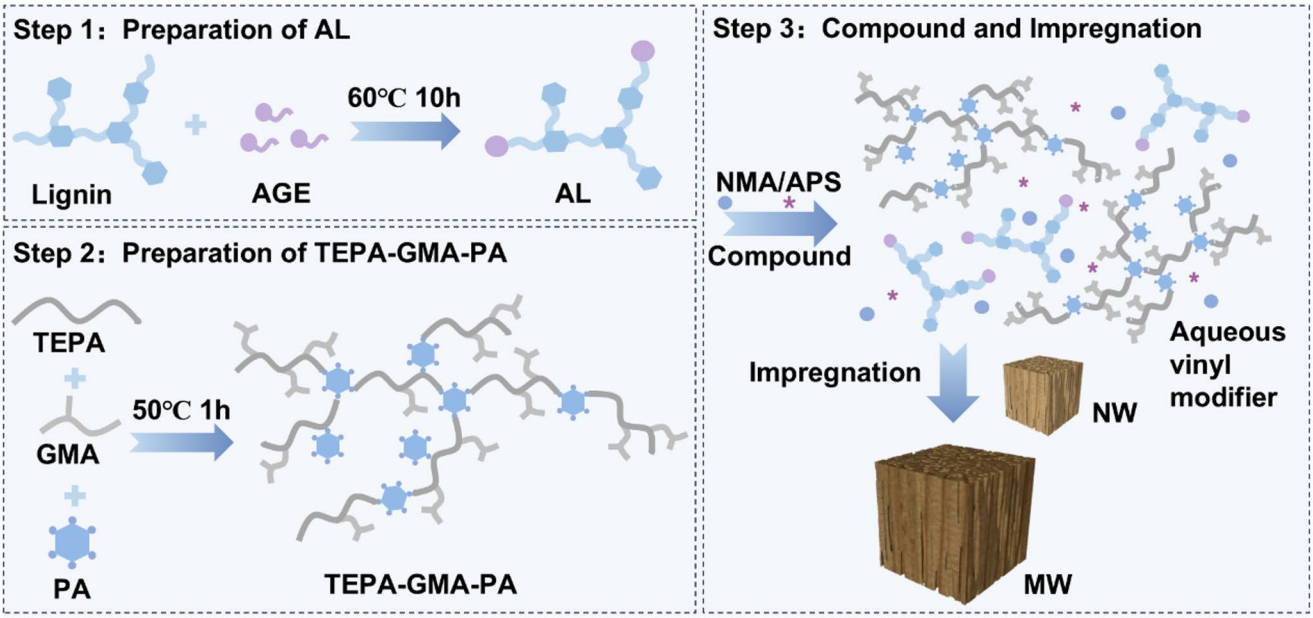

Fig. 1 Schematic illustration of fabrication of MW

was purchased from Tianli (Tianjin, China). All chemicals were used without further purification.

2.2 Preparation of allylated lignin

The purification of lignin was carried out following the method described in a previous study (Yu et al. 2023). Purified lignin (10 g) and NaOH (0.1 mol, 4 g) were dissolved in distilled water (10.88 mol, 196 g). After complete dissolution, AGE (0.114 mol, 13 g) was added drop by drop using a springe. The reaction proceeded at 60 °C for 10 h, subsequently the reaction system was cooled to room temperature and washed three times by acid precipitation (1.316 mol, 400 g, 12 wt%) and alkali dissolution (0.1 mol 40 g, 10 wt%). The obtained allylated lignin (AL) was concentrated to a 15 wt% solution and stored at low temperature for further use.

2.3 Preparation of unsaturated phytic acid amine

PA (0.005 mol, 4.7 g, 70 wt%) was diluted with distilled water (2.187 mol, 39.4 g) and set aside. TEPA (0.03 mol, 5.67 g) and GMA (0.06 mol, 8.52 g) were introduced into a 100 ml three-neck flask equipped with a mechanical stirrer and a condenser. The mixture was heated to 50°C and maintained for 20 min, after which the prepared PA solution was gradually added dropwise in small quantities for 1 h. Upon completion, the reaction was cooled to room temperature, and the mixture was poured into 1 L of anhydrous ethanol, resulting in the precipitation of product. The precipitation was collected and repeatedly dissolved in water, followed by reprecipitation in anhydrous ethanol (10 times the volume of water). The resulting product was dried to yield purified unsaturated phytic acid amine named TEPA-GMA-PA. And the solution resulting from the reaction between TEPA and GMA was washed with anhydrous ethanol and dried, affording the product TEPA-GMA.

2.4 Preparation of modified wood

Different lignin-based modifiers were prepared according to Table 1. Fir samples were impregnated under vacuum at −0.1 MPa for 2 h, followed by pressure impregnation at 0.7 MPa for 12 h, ensuring complete submersion in the modifier throughout the process. After impregnation, the sample surfaces were wiped with a nonwoven fabric and air-dry for 2 days. A stepwise drying process was the applied: drying at 40 °C and 80 °C for 4 h each, followed by curing at 103 °C for 12 h. The final modified fir samples were obtained and named as specified in Table 1.

2.5 Characterization

The chemical bond or functional group in the specimens were analyzed using Fourier-transform infrared spectrometry (FTIR, Nicolet 6700; Thermo Fisher Scientific, Waltham, MA, USA). The FTIR spectra of dried samples were obtained within the range of 4000–400 cm^{-1} at 32 scans/min. The molecular structures of samples were determined using a nuclear magnetic resonance spectrometer (NMR, Avance III HD 500 MHz; Bruker, Switzerland). For ^{1}H NMR and ^{13}C NMR analyses, a 150 mg sample of AGE, GMA, lignin, allylated lignin specimens dissolved in 1 mL of DMSO-d_6 was subjected to 64 scans and 2048 scans, respectively. A 150 mg sample of TEPA, TEPA-GMA () and TEPA-GMA-PA specimens dissolved in 1 mL of D_2O was subjected to 64 scans and 2048 scans, respectively. The morphology and structure of the specimens were characterized by scanning electron microscopy (SEM; CXS-5TAH-119402; COXEM, Daejeon, Korea) at an accelerating voltage of 15 kV. Additionally, energy-dispersive spectroscopy (EDS) was utilized to study the elemental distribution on the cross-section of the specimens. All the specimens were sectioned from the interior of 10 mm × 20 mm × 20 mm (L × T × R) wood blocks and were coated with Au before observation. Elemental changes within the specimens were examined through X-ray photoelectron spectroscopy (XPS; ESCALAB 250Xi, Thermo Fisher Scientific, USA). The thermal degradation behavior of specimens was analyzed using a Thermal gravimetric analyzer (TG, TGA-4000, PerkinElmer, USA). Specimens (5–10 mg) were heated from room temperature to 800 °C at a rate of 20 °C/min under N_2 atmosphere. The combustion behavior of specimens with dimensions of 5 mm × 100 mm × 100 mm (Longitudinal × Tangential × Radial directions) was evaluated using a cone calorimeter (CCT, CONE 6810, Suzhou, Yangyi Vouch Testing Technology Co., Ltd., China), under a 35 $KW m^{-2}$. Each group underwent testing in accordance with GB/T 34,749-2017. The limiting oxygen indices of specimens with dimensions of 80 mm × 5 mm × 3.2 mm

Table 1 Names of modified wood specimens with different lignin-based flame retardants

Labels	TEPA-GMA-PA (g)	NMA (g)	AL (g)	APS (g)	H_2O (g)	Total (g)	Solid content (%)
NW	-	-	-	-	30	30	0
MW/TN	3	2	0	0.05	24.95	30	16.8%
MW/TNL-1	3	2	0.25	0.053	24.70	30	17.7%
MW/TNL-2	3	2	0.5	0.055	24.45	30	18.5%
MW/TNL-3	3	2	0.75	0.075	24.18	30	19.4%

(Longitudinal × Tangential × Radial directions) were determined using an oxygen index apparatus (LOI JF-3, Jiangning Analytical Instrument Company, China), in accordance with GB/T 2406.2-2009 standards.

2.6 Physical and mechanical properties

The specimens with dimensions of were used for weight and dimensional stability tests, with 8 replicates in each group. The weight% gain (WPG), Bulking values and Density were given by the following equations:

$$\text{WPG}\ (\%) = \frac{\text{M1} - \text{M0}}{\text{M0}} \times 100\% \tag{1}$$

$$\text{Bulking}\ (\%) = \frac{\text{V1} - \text{V0}}{\text{V0}} \times 100\% \tag{2}$$

$$\text{Density}\ (\%) = \frac{\text{M1}}{\text{V1}} \times 100\% \tag{3}$$

Here, M0 is the oven-dried weight of specimens before modification, M1 is the oven-dried weight of specimens after modification, V0 is the oven-dried volume of specimens before modification, and V1 is the oven-dried volume of specimens after modification.

In order to assess the dimensional stability of modified wood and the leaching rate of the modifier, the specimens underwent one cycle of leaching. The specimens were submerged in deionized water and changed once a day until the specimen mass was constant (about 7 days). Subsequently, the specimens were air-dried for 1 week and continuously dried in an oven at 40 °C, 80 °C and 103 °C for 4 h. The mass and dimensions of the specimens were measured in water-saturated and oven-dried conditions, respectively. And the swelling coefficient (SC), anti-swelling efficiency (ASE), Water absorption rate (WAR) and leachability of modifier (Leaching) were given by the following equations:

$$\text{SC}\ (\%) = \frac{\text{V3} - \text{V2}}{\text{V2}} \times 100\% \tag{4}$$

$$\text{ASE}\ (\%) = \frac{\text{SC1} - \text{SC2}}{\text{SC1}} \times 100\% \tag{5}$$

$$\text{WAR}\ (\%) = \frac{\text{M2} - \text{M1}}{\text{M1}} \times 100\% \tag{6}$$

$$\text{LM}\ (\%) = \frac{\text{M1} - \text{M3}}{\text{M1} - \text{M0}} \times 100\% \tag{7}$$

Here, V2 is the volume of oven-dried specimens before leaching, V3 is the volume of water-saturated specimens, SC1 is the swelling coefficient of unmodified specimens, SC2 is the swelling coefficient of modified specimens, M2 is the weight of water-saturated specimens, and M3 is the oven-dried weight of specimens after leaching.

The specimens with dimensions of 180 mm × 10 mm × 10 mm (Longitudinal × Tangential × Radial directions) were used for bending strength tests, with 8 replicates in each group. Bending properties were performed using a three-point bending test with a testing speed of 5 mm min^{-1} with a span of 150 mm (C41.103, Ssans Shanghai, China). The specimens with dimensions of 135 mm × 13 mm × 13 mm (Longitudinal × Tangential × Radial directions) were used for impact strength tests, with 8 replicates in each group. Impact strength was measured using a 15 J shaker (XJC-25; Chengde, Chengde, China).

3 Results and discussion

3.1 Chemical structure analysis of alkylated lignin

AL was synthesized via an etherification reaction, which is shown schematically in Fig. 2a. In addition to the formation of ether bonds in the lignin molecule, a new hydroxy group is generated in the side chain, which can further participate in the etherification modification as an active site for the subsequent reaction. The occurrence of the etherification reaction was confirmed through FTIR and NMR analysis. In the FTIR spectra, the broad -OH stretching vibration band around 3400 cm^{-1} persisted due to the formation of new hydroxyl groups following epoxy ring-opening of AGE. Concurrently, the characteristic absorption band of the epoxide ring in AGE at approximately 910 cm^{-1} disappeared in the spectrum of AL, confirming the consumption of the epoxy group via the ring-opening reaction. Compared to lignin, AL exhibited enhanced C-H stretching vibrations at 2851 cm^{-1}, corresponding to methyl and methylene groups, and the disappearance of the phenolic -OH absorption at 1361 cm^{-1}, indicating the successful reaction between AGE and lignin phenolic hydroxyls (Sohni et al. 2019). Additionally, a new absorption band at 924 cm^{-1}, attributed to the -CH bending vibration of RCH=RCH, further confirmed the allyl group was successfully grafted inlignin (Younesi-Kordkheili and Pizzi 2021). In the ^{1}H NMR spectra (Fig. 2c, d), the resonance peaks of a-c were assigned to the protons in the epoxide ring of AGE. The disappearance of these signals in the spectrum of AL indicates the opening of the epoxide ring and its subsequent grafting onto lignin during the etherification reaction. The peaks at 5.8, 5.2 and 5.1 ppm (g_1, h_1, i_1) corresponded to the vinyl protons in RCH=RCH structures (Wang et al. 2023b; Jin et al. 2018). The resonance peaks of k, j_1 and l were assigned to methoxy protons and protons on the grafted side chains. At the same time,

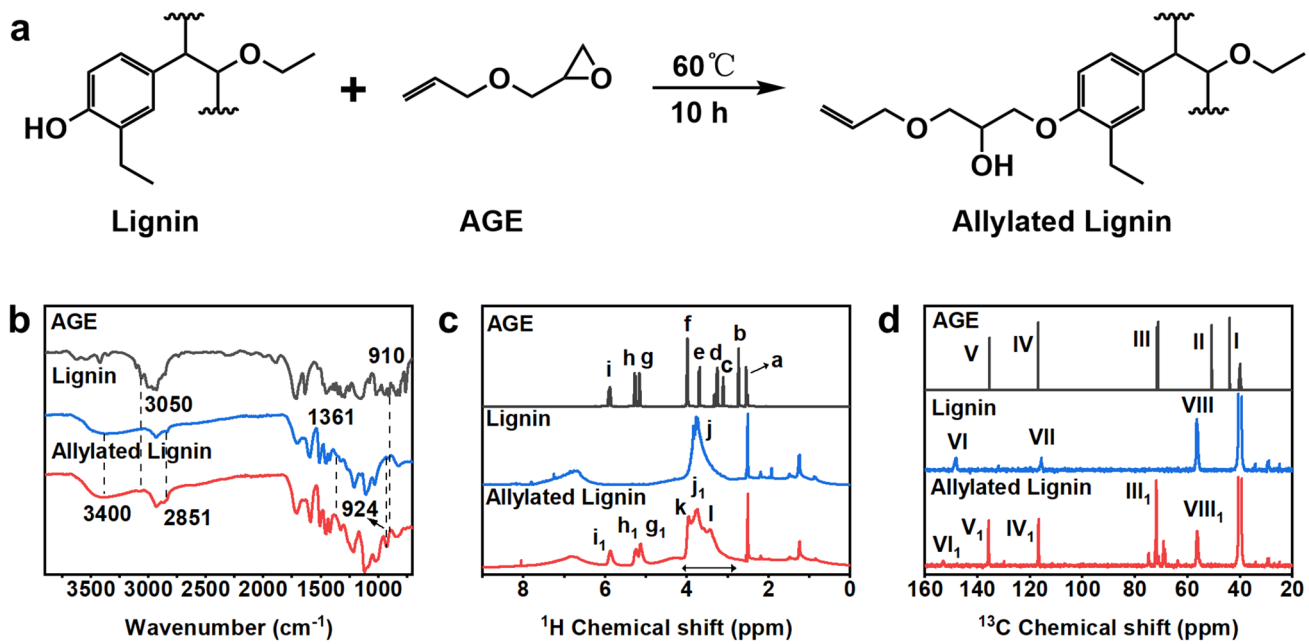

Fig. 2 Schematic illustration of fabrication of AL (**a**). FTIR (**b**), ^{1}H NMR (**c**) and ^{13}C NMR (**d**) spectra of lignin and AL

peak d was likely a residual water peak from the AGE sample. Further evidence was provided by ^{13}C NMR spectra. AL retained the characteristic double bond carbon signals at 135.8 and 116.5 ppm (IV_1, V_1) from its structure. The peaks at 115.4 ppm (IV_1) at AL attributed to the etherified C5 of the guaiacyl ring. Crucially, the disappearance of the epoxide ring carbon signals (I, II) of AGE in the AL spectrum collectively demonstrates that the epoxide ring of AGE underwent a ring-opening reaction with the hydroxyl groups of lignin. This resulted in the grafting of the allyl ether chain onto the lignin backbone.

3.2 Chemical structure analysis of TEPA-GMA-PA

The unsaturated phytic acid amine was synthesized through a two-step reaction, as illustrated in Fig. 3a. Initially, GMA underwent a ring-opening addition with TEPA to yield TEPA-GMA. Subsequently, PA solution was added to neutralize the excess amino groups, resulting in the formation of the final product, TEPA-GMA-PA.

The successful synthesis of TEPA-GMA-PA was confirmed by ^{1}H NMR spectroscopy. As shown in Fig. 3b and c, the peaks at 6.1 and 5.8 ppm correspond to the vinyl proton signals of GMA (peaks I and II). The persistence of these peaks in the spectra of both TEPA-GMA and TEPA-GMA-PA confirmed that the C=C double bond remained intact throughout the reaction process (Jiang et al. 2025; Wang et al. 2022). Simultaneously, the disappearance of the proton signals of the epoxide ring in GMA (peaks V, VI, and VIII at 3.2, 2.8, and 2.6 ppm, respectively) in the TEPA-GMA spectrum demonstrated the occurrence of the ring-opening reaction between the epoxy group of GMA and the amine groups of TEPA (Wang et al. 2024; Yang et al. 2023a). These characteristic signals were present in both TEPA-GMA and TEPA-GMA-PA, confirming the successful grafting of GMA onto the TEPA backbone. Upon the addition of PA, its phosphate groups underwent a neutralization reaction with the amine groups, simultaneously enhancing intra- and intermolecular hydrogen bonding. The formation of hydrogen bonds reduced the electron density around the protons, leading to a stronger de-shielding effect. This was manifested by the altered peak shape of the methylene proton signal at 2.6 ppm (in TEPA-GMA) and its shift toward a higher chemical displacement. These observations confirm the successful formation of the ionic complex TEPA-GMA-PA via the reaction between TEPA-GMA and phytic acid.

3.3 Physical properties and mechanical performance of wood

The comparison of physical properties of natural wood and modified wood were shown in Fig. 4. The penetration of the modifier into wood was influenced by both wood structural characteristics and physicochemical properties of the modifier, such as viscosity and polarity. In this study, the vacuum-pressure impregnation process, combined with the low-viscosity aqueous modifier, promoted effective penetration through the wood. The high compatibility of the modifier with the hydrophilic wood cell walls further enhanced this process. The WPG results demonstrated the good penetration within the wood. With the addition of AL, the WPG of the modified wood increased from 29.1

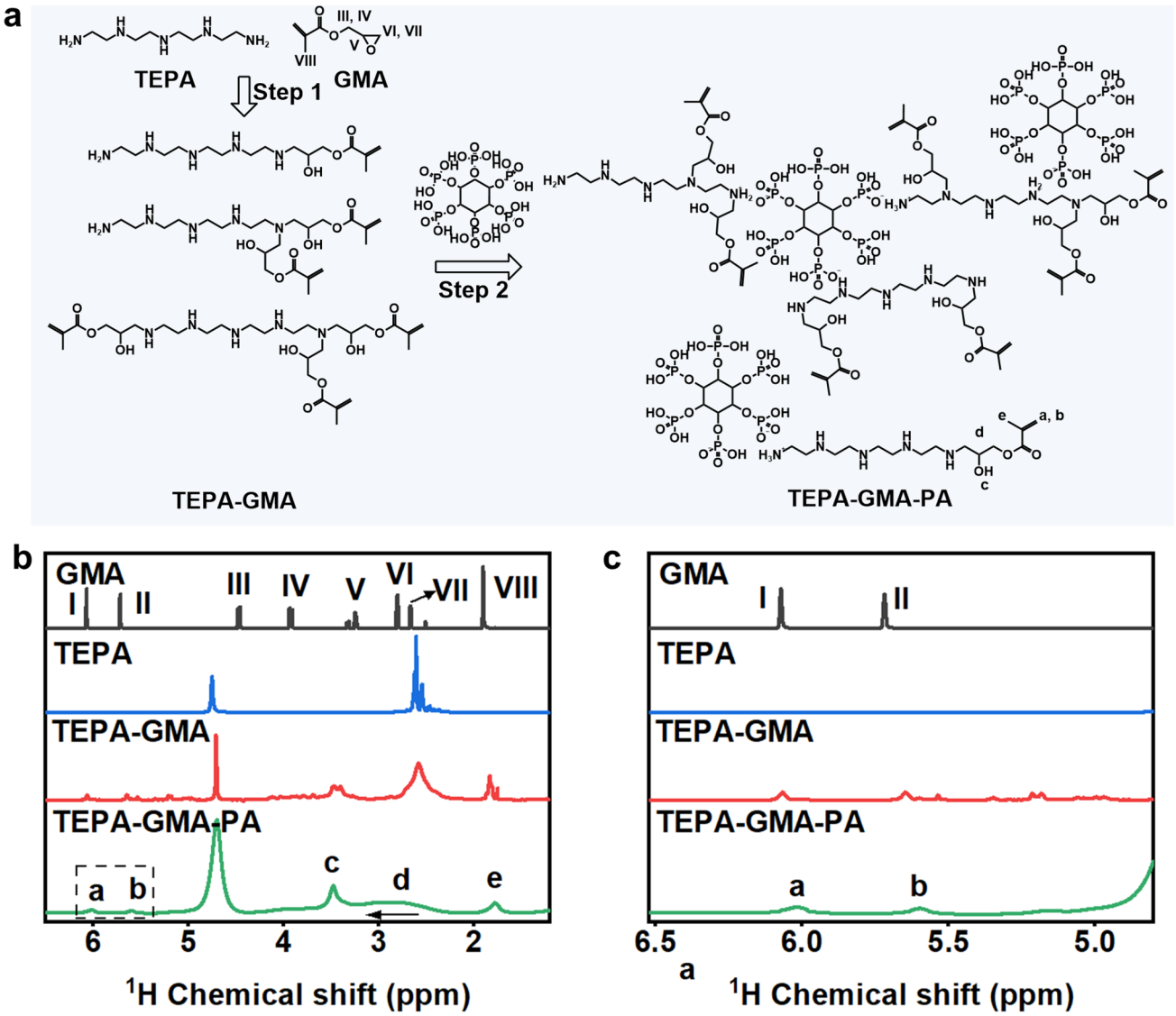

Fig. 3 Schematic illustration of fabrication of TEPA-GMA-PA (**a**). ^{1}H NMR of TEPA and TEPA-GMA-PA (**b**, **c**)

to 43.7%, attributed to the higher solid content of the modifier depositing within the wood structure upon curing (Fig. 4a). As shown in Fig. 4b, the bulking of modified wood ranged from 4.3 to 4.9%. The increasing content of AL did not significantly affect the diffusion of the low-molecular-weight monomer NMA within the wood. Overall, the oven-dry density of the specimens increased from 0.32 g/cm^3 in natural wood to 0.45 g/cm^3 in MW/TNL-3, indicating that the higher density imparted by the modifier contributes to improved mechanical properties of the modified wood (Fig. 4c). As a typical porous material, wood contains cell lumens, pits and micropores in the cell walls, which serve as pathways and spaces for water NW exhibited a WAR of 251%, all modified samples showed significantly reduced water uptake. Specifically, MW/TNL-3 exhibited a decreased WAR of 152% (Fig. 4d), which was closely associated with the reduction in pore volume. The modifier polymerized in situ within the cell lumen and cell wall to form a continuous structure, which effectively filled these pores and obstructed the water transport channels, leading to the reduction of water absorption. Water absorption and swelling of the cell wall were the main cause of volume changes in wood, within a certain range, the effect of modifiers on cell wall filling showed a positive correlation with their enhancement of wood dimensional stability. As a result of the bulking effect in cell wall, the dimensional stability of the modified wood specimens was improved (Ermeydan et al. 2021; Guo et al. 2022). As shown in Fig. 4e, the ASE of MW/TN was 43.3%, the increase in the addition of AL did not significantly affect the dimensional stability of the modified wood, which was consistent with the results of the volume increase rate. The lignin-based composite modifier

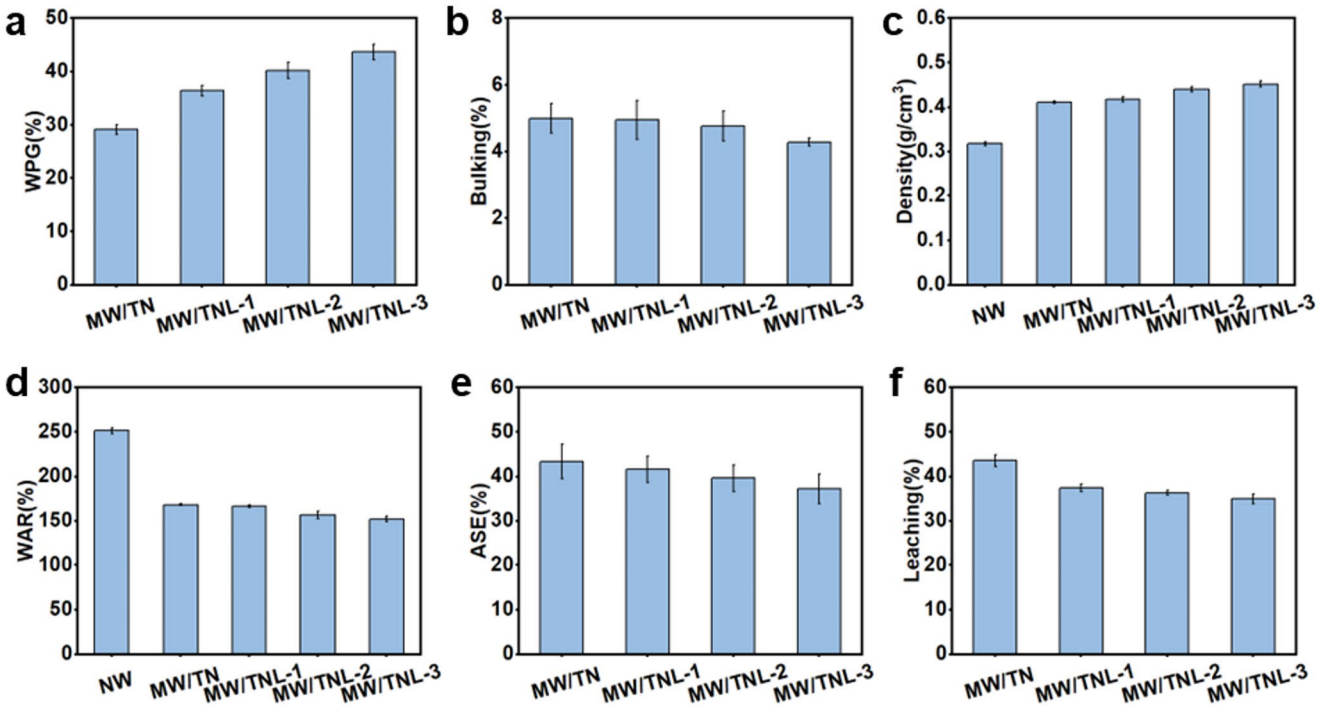

Fig. 4 WPG (**a**), bulking (**b**), density (**c**), WAR (**d**), ASE (**e**) and leaching of modified wood

with reactive double bonds formed a water-insoluble polymer after curing inside the modified wood, and the leaching of MW/TN was 43.6%. The addition of AL further copolymerized with the TEPA-GMA-PA and NMA, which reduced the MW/TNL-3 leaching to 34.9%.

Combined with the SEM images in Fig. 5, the distribution of the modifiers within the modified wood could be observed. In the natural wood (Fig. 5a1-a4), the cell wall appeared thin, with large tracheid lumens. The vertical sections revealed smooth cell wall surfaces and well-connected pits, indicating high permeability. In the MW/TN without AL, the tracheid lumens remained largely unfilled. Elemental mapping of N and P revealed distributions corresponding similar to the shape of the cell wall, suggesting that the modifier either penetrated into or was deposited along the inner surfaces of cell wall (Fig. 5b2-b3). Combined with the analysis of WPG and ASE, AL was difficult to impregnate into the cell wall due to its large molecular size, which was also confirmed with the filling of cell lumen by polymer in Fig. 5c1-e1 and Fig. 5c4-e4, the filling was relatively dispersed. The polymers exhibited good interfacial adhesion with the cell walls, forming tight and continuous contact, which provides structural support for improved dimensional stability and mechanical performance. For MW/TN specimen, the N and P elements were uniformly distributed in a pattern matching the cell wall structure, indicating that the flame retardant coated or infiltrated the cell walls. Furthermore, N and P elements were also observed within the cell lumens of MW/TNL specimens. This distribution can be attributed to the AL, which can polymerize with NMA and TEPA-GMA-PA, thereby obstructing the cell lumens and resulting in the concomitant presence of the flame-retardant elements therein. These polymers also obstructed numerous pits on the cell walls, effectively delaying moisture migration into the cell wall, thereby enhancing the water resistance and stability of the modified wood (Fig. 5c1-e1, Fig. 5c4-e4).

The bending performance, modulus of rupture (MOR), modulus of elasticity (MOE) of natural wood and modified wood are shown in Fig. 6a. The MOR of MW/TN increased from 72.5 to 87.6 MPa and the MOE increased from 10.7 to 12.5 GPa compared with the natural wood. The enhancement in bending performance was positively correlated with the addition of AL. Specifically, MW/TNL-3, with the highest amount of AL, exhibited a MOR of 94.8 MPa and MOE of 12.8 GPa, representing increases of 30.8% and 19.6%, respectively, compared to natural wood. This enhancement in bending properties could be attributed to the reinforcing effect of the polymerization network in wood (Guo et al. 2022; Wang et al. 2023a). The firm polymer filling in the cell lumens and the strong interfacial bonding with the cell walls facilitate efficient transfer of external loads to adjacent cell wall or polymer phases, mitigating stress concentration and resulting in enhanced macroscopic mechanical performance. Impact strength, an important index for evaluating the toughness of materials, showed no significant change

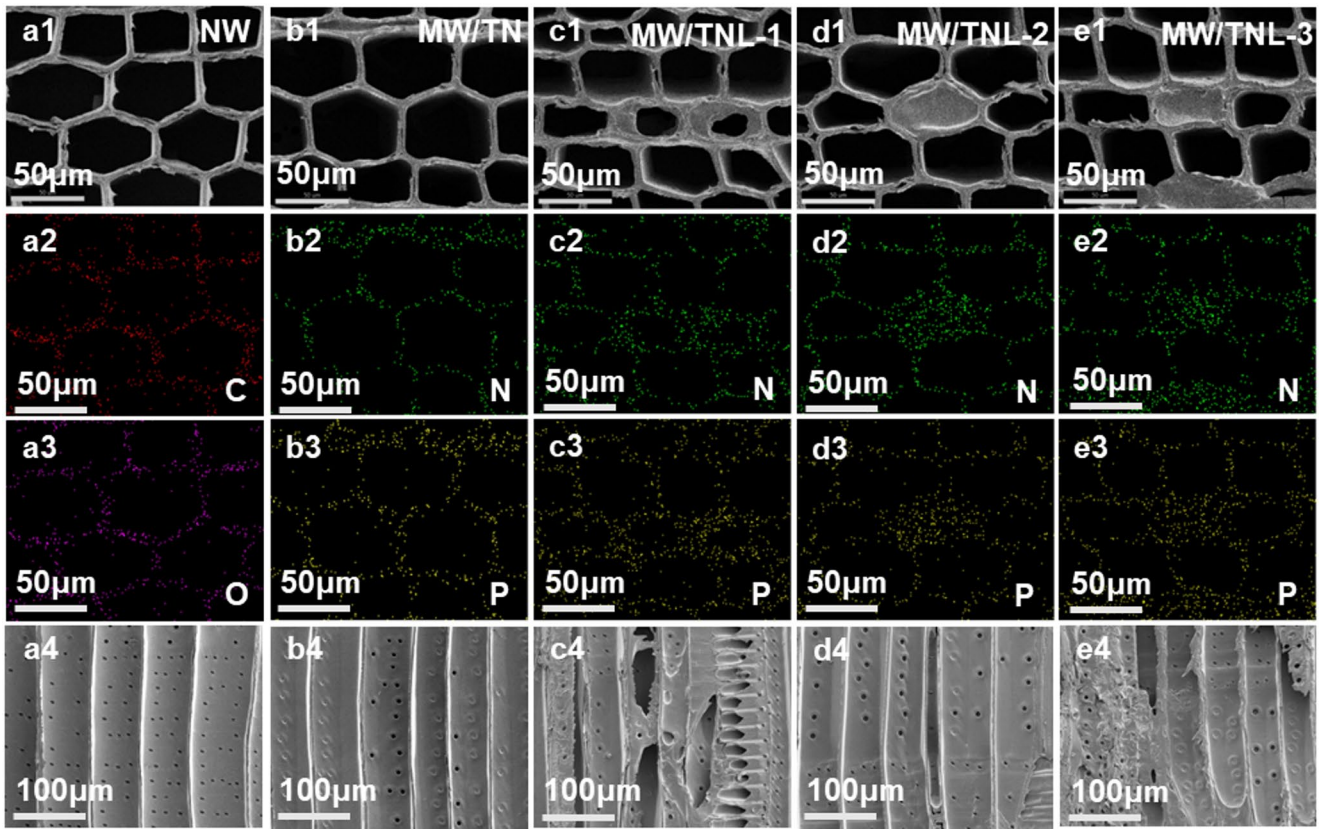

Fig. 5 SEM images of the transverse section of natural wood and modified wood (**a1-e1**), elemental maps of C, O of natural wood (**a2**, **a3**), elemental maps of N, P of modified wood (**b2-e2**, **b3-e3**) and SEM images of vertical section of natural wood and modified wood (**a4-e4**)

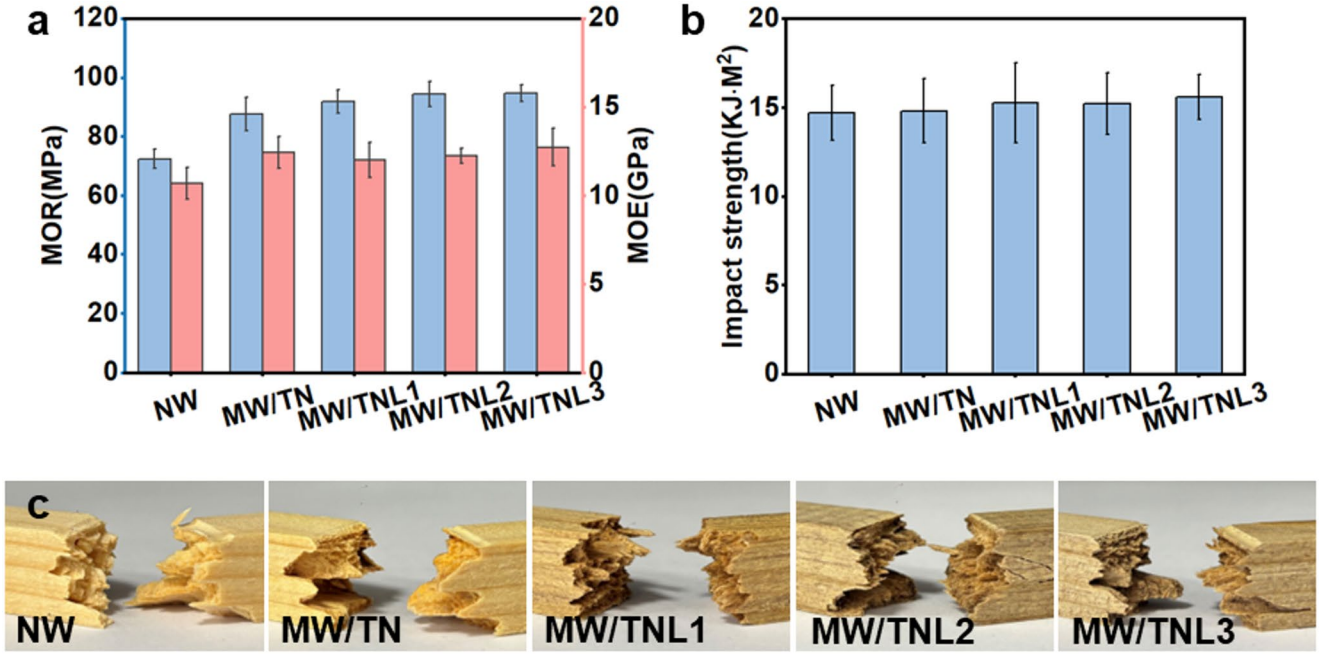

Fig. 6 MOR, MOE (**a**), impact strength (**b**) of NW and modified wood and the photographs of the section after impact test for NW and modified wood (**c**)

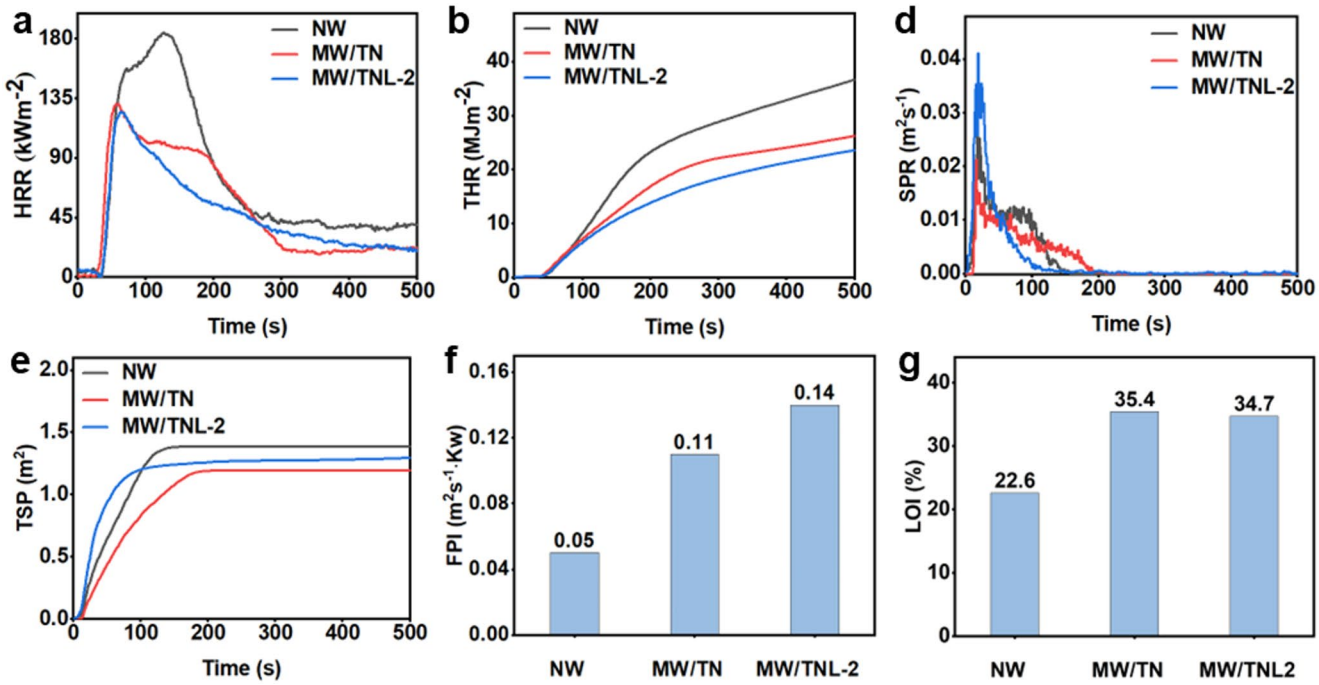

Fig. 7 HRR curves (**a**), THR curves (**b**), SPR curves (**c**) TSP curves (**d**) FPI value and LOI value of NW and modified wood

Table 2 Cone calorimetry test data of natural wood and modified wood

Labels	TTI (s)	P_1HRR (kW m^{-2})	P_2HRR (kW m^{-2})	THR (MJ m^{-2})	P_1SPR (m^{-2} s^{-1})	P_2SPR (m^{-2} s^{-1})	TSP (m^2)
NW	10	157.7	184.8	36.8	0.0269	0.0127	1.39
MW/TN	14	130.1	102.8	26.3	0.0213	-	1.19
MW/TNL-2	18	124.7	-	23.7	0.0411	-	1.29

of all specimens (Fig. 6b). This retention of the toughness of the wood itself may be related to several reasons, lignin-based modifier pH mild to avoid degradation of the wood components. The modifiers do not react with the wood cell wall, ensuring that the cell wall microfibrils slipped and absorbed energy through deformation (Zhou et al. 2024). The interwoven long-chain structures within the polymer network effectively dissipated energy upon impact (Sun et al. 2016). Figure 6c showed sectional images of specimens following impact tests. Fracture surface analysis after impact loading revealed typical features of ductile fracture. The modified specimens displayed tearing and fiber pull-out similar to those observed in natural wood, resulting from relative sliding between fibers during impact. This was caused by the relative slip between fibers when subjected to impact loading, which further indicated that the double-bond modifier system has no significant effect on the toughness of modified wood.

3.4 Combustibility of wood

The cone calorimeter can simultaneously obtain the dynamic change information of heat. smoke and mass during the combustion process, simulate the real combustion environment, be used to characterize the combustion performance of materials (Wang et al. 2017). Figure 7; Table 2 showed the time to ignition (TTI) heat release rete (HRR), total heat release (THR), smoke production rate (SPR) and total smoke production (TSP) of natural wood and modified wood. The TTI of the natural wood is 10 s, the TTI of MW/TN and MW/TNL-2 were extended to 14 s and 18 s, respectively, which could allow reaction time in case of fire. The HRR cure of the natural wood displayed typical characteristic peaks of lignocellulosic combustion. The first peak (P_1HRR) occurred due to rapid ignition under thermal radiation, following by char layer formation on the wood surface, which suppressed further heat release, the flammable gases generated broke through the charcoal layer, resulting in the second heat release rate peak (P_2HRR) (Zhou et al. 2023). The HRR of modified wood decreased after the addition of modifier, in which the P_1HRR and P_2HRR of MW/TN decreased by 17.5% and 44.4%, suggesting that the modifier can effectively mitigate the spread of the fire. MW/TNL-2 showed the first peak of HRR and had a reduction of 20.9%. The decrease in HRR of the modified wood also led to the reduction in THR, the THR of MW/TN and

MW/TNL-2 were 26.3 $kW \cdot m^{-2}$ and 23.7 $kW m^{-2}$, which were 28.5% and 35.6% lower than that of the natural wood (36.8 $kW m^{-2}$). As shown in Fig. 7b, THR increased rapidly during the flaming combustion stage (between ignition and P_2HRR), reaching the highest growth rate at the P_2HRR. Afterward, the rate of increase in THR showed considerably, indicating that the majority of heat was released during flaming combustion. It can be attributed to the dehydration charring of the wood promoted by PA, which forms a char layer that prevents the transfer of heat and combustible gases (Shen et al. 2025; Pan et al. 2024). These fundings highlight the critical importance of suppressing flaming combustion to improve the flame retardancy of wood.

Smoke particles generated during wood combustion consist of a mixture of solid, liquid, gaseous components. Figure 7d and e present the SPR and TSP curves of natural wood and modified wood. Similar to the HRR, the SPR of natural wood appeared two peaks, however, the first smoke release rate peak (P_1SPR) appeared early in the ignition stage was significantly higher than the second peak (P_2SPR), with values of 0.0269 $m^2 s^{-1}$ and 0.0127 $m^2 s^{-1}$, respectively. The corresponding TSP reached 1.39 m^2, indicating considerable smoke generation during the initial flaming phase. In contrast, all modified wood specimens occurred a single peak in SPR curves, suggesting that the modification suppressed continuous smoke generation in later combustion stages. The P_1SPR of the MW/TN specimen remained comparable to that of the natural wood, while MW/TNL-2 showed a higher P_1SPR of 0.0411 $m^2 s^{-1}$. The increase may be attributed to the presence of a greater proportion of non-combustible volatiles such as water vapor, or incomplete oxidation of AL due to lower combustion temperatures and relatively oxygen-deficient conditions (Zhang et al. 2023b). Despite the elevated initial SPR in MW/TNL-2, the TSP values of the modified wood specimens were reduced. The TSP for MW/TN and MW/TNL-2 were 1.19 m^2 and 1.29 m^2. The reduction was associated with the formation of a dense char layer during combustion, which acted as an effective barrier and absorbent, limiting the release of smoke particulates.

The fire performance index (FPI) was a property of the material itself, calculated as the ratio of TTI and PHRR (Zhang et al. 2023a). The FPI was independent of the mass and volume of the material, it reflected the thermal response behavior of the material, with higher values indicating lower fire hazard. As shown in Fig. 7f, The FPI of the natural wood was 0.05 $m^2 s kW^{-1}$, while the FPI of MW/TN and MW/TNL-2 were 0.11 $m^2 s kW^{-1}$and 0.14 $m^2 s kW^{-1}$, which were 120% and 180% higher than that of the natural wood. These results showed that the combined flame-retardant system composed of TEPA-GMA-PA and NMA effectively enhanced the fire resistance of natural wood, the additional incorporation of AL further reduced the fire risk. The limiting oxygen index (LOI) measures the minimum oxygen concentration in an oxygen/nitrogen mixture required to support the combustion of material. A higher LOI value corresponded to greater flame retardancy. The LOI of the natural wood was 22.6%, classing it as a flammable material, the LOI of MW/TN increased to 35.4%. Although the incorporation of AL caused a slight reduction in LOI, which may due to the relatively high calorific value of lignin, the LOI of MW/TNL-2 was 34.7%, indicating that the modified wood could be categorized as flame-retardant materials.

3.5 Flame-retardant mechanism analysis

The macroscopic and microscopic morphologies, as well as the elemental distribution of the specimens before and after CCT tests, were shown in Fig. 8a. The surface of MW/TN and MW/TNL-2 modified specimens appeared slightly darker than that of the natural wood, while the characteristic wood grain remained clearly visible. After CCT tests, the natural wood was burned through after the test, the structure was loosened, with the surface covered with grayish-white ash. In contrast, MW/TN and MW/TNL-2 exhibited intact char layers with irregular and uneven cracks on the surface. Notably, the MW/TNL-2 showed wider surface fissures, which may have facilitated the release of smoke particles and contributed to its higher peak smoke production rate. Nevertheless, the continuous char layer still played a critical role in thermal insulation and oxygen barrier during combustion (Lai et al. 2025; Wang et al. 2015; Yuan et al. 2024). SEM-EDS analysis was employed to investigate the distribution of the modifiers within the residual char. After combustion, the filler within the cell lumens disappeared, while N and P elements remained in the cell wall of MW/TNL-2. This indicated that the polymer participated in the rehydration and carbonization of wood components or AL during pyrolysis, thereby promoting the formation of a stable char layer.

The pyrolysis of wood was a complex process involving both physical and chemical changes. Figure 8b illustrated the mass loss and temperature relationship of natural wood and modified wood during thermal degradation under a nitrogen atmosphere, with corresponding thermal decomposition parameters listed in Table 3. Both natural wood and modified wood specimens exhibited similar thermal degradation behavior. For natural wood, the major decomposition occurred within the range of 240–400 °C, with the maximum peak temperature of thermal decomposition rate (T_{max}) of 372 °C. During the stage, chemical constituents such as cellulose and lignin underwent dehydration and decarboxylation reactions, releasing H_2O, CO_2 and a large quantity of flammable gases (Yan et al. 2023; Yang et al. 2023b). As the decomposition proceeded, the rate gradually

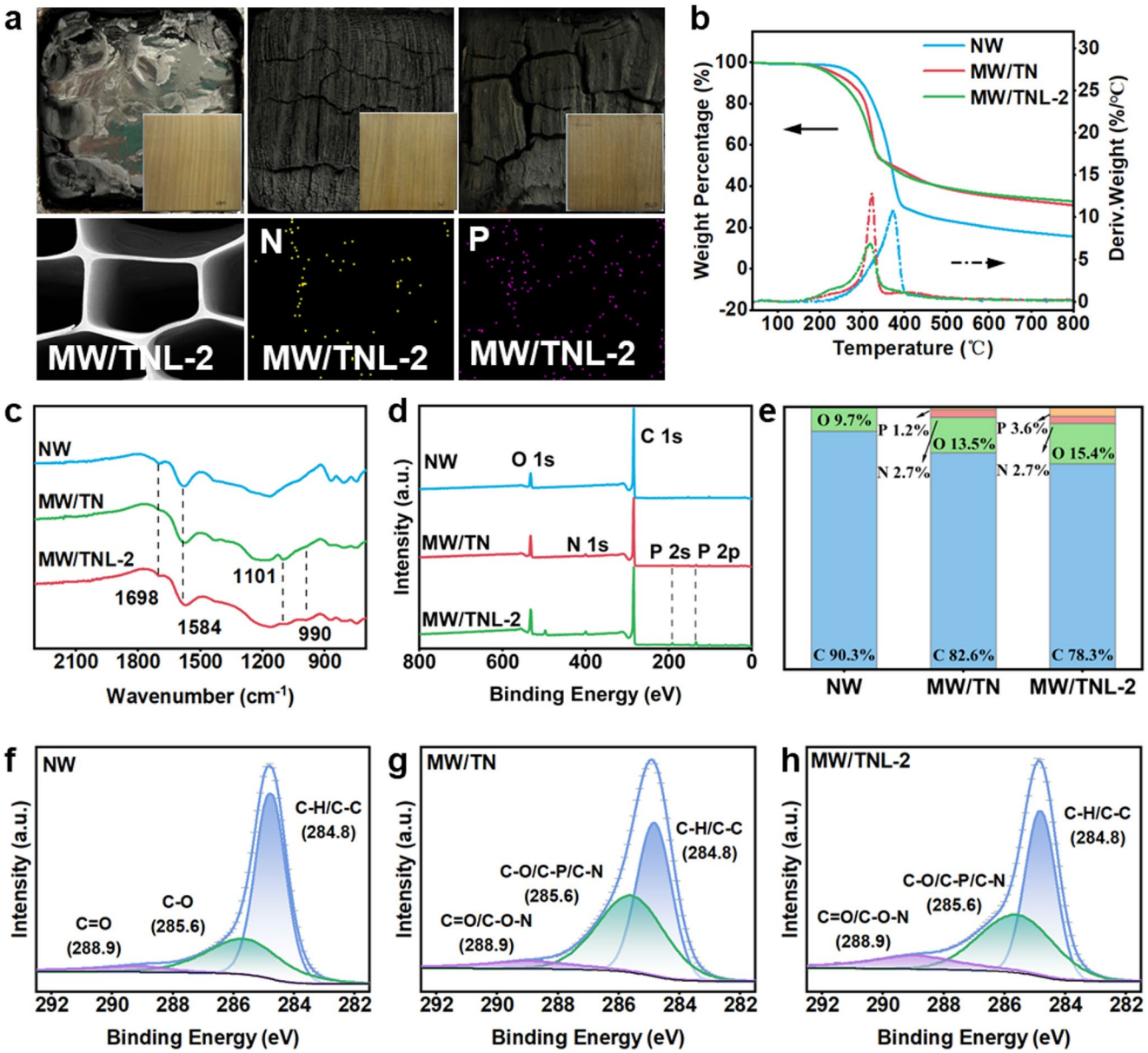

Fig. 8 The macroscopic, microscopic structures of fir and modified fir before and after CCT, SEM and elemental maps of N, P of fir and modified fir after CCT (**d**, **e**), TG and DTG spectra of natural wood and modified wood (**b**), FTIR spectra and XPS spectra of the char residue in the fir and modified fir after CCT (**c**, **d**), elemental compositions of the char residue in the fir and modified fir after CCT (**e**) high-resolution C 1 s spectra of the char residue in the fir (**f**) and modified fir (**g**, **h**)

Table 3 TG and DTG data of natural wood and modified wood

Labels	$T_{5\%}$ (°C)	$T_{50\%}$ (°C)	T_{max} (°C)	R_{max} (%/°C)	Residual charcoal rate (%)
NW	274	368	372	10.7	15.5
MW/TN	232	372	323	12.8	30.6
MW/TNL-1	222	350	324	7.9	31.0
MW/TNL-2	218	363	317	6.9	32.6
MW/TNL-3	224	357	326	7.3	31.6

declined, charcoal further underwent deoxygenation and dehydrogenation in the form of CO and formaldehyde, followed by aromatization. Ultimately, the char yield at 800 °C was 15.5%. Compared to natural wood, the modified wood exhibited a significantly earlier onset of decomposition. The initial decomposition temperature ($T_{5\%}$) and T_{max} of MW/TNL-2 were 218 °C and 317 °C, respectively, while the peak of maximum rate of mass loss (R_{max}) was reduced to 6.9%/°C, the residual charcoal rate was increased to 32.6%. The results indicated that the PA component of modifier

acted as an acid catalyst to promote the rewatering of the wood into charcoal.

The chemical composition of the residual char was further investigated using FTIR and XPS, as shown in Fig. 8c and d. In the residual char of the natural wood, the peaks at 1704 cm^{-1} and 1580 cm^{-1} correspond to the stretching vibration of C=O and the skeletal vibration of aromatic rings (Yan et al. 2025b). In contrast, new absorption peaks at 1093 cm^{-1} and 988 cm^{-1} appeared in the residual char of MW/TN and MW/TNL-2, which were attributed to P-N and P-O-C structures (Fan et al. 2025). This demonstrated that the flame retardant participated in the carbonization process of AL during pyrolysis, contributing to the formation of a stable char network. XPS analysis further revealed the elemental composition of the residual char (Fig. 8d and e). The elemental contents of N and P in the residual char of MW/TN were 2.7% and 1.2%. With the introduction of AL, more P element was retained in the char, increasing the P content to 3.6%. High resolution C 1 s spectra of the char residues showed three distinct peaks. The covalent bonds corresponding to C-O/C-P/C-N at 285.6 eV were more pronounced in MW/TN and MW/TNL-2 than in that in natural wood (Xie et al. 2024). Additionally, due to the presence of carbonyl groups in AL, the intensity of the peak at 288.9 eV, representing C=O bonds, was further enhanced (Yan et al. 2025a). This suggested that AL promotes the cross-linking within the char layer network, the introduction of higher bond energy structures enhanced the chemical stability of the resulting char.

4 Conclusion

In conclusion, a novel aqueous lignin-based wood modifier was synthesized to enhance the dimensional stability, mechanical strength and flame retardancy of fast-growing wood. The modifier underwent in-situ polymerization within the wood structure, forming a three-dimensional crosslinked network in cell wall and cell lumen. Covalent bonding facilitates strong fixation of modifier and enhances the dimensional stability of the modified wood, with modified wood achieving an ASE of 41.6%, a leaching of 34.9%. The dense polymer filling and its tight interfacial bonding with the cell wall enable efficient load transfer and stress dispersion. As a result, modified wood exhibited a 30.8% increase in MOR and 19.6% in MOE, without significantly affecting impact strength. The modified wood demonstrated good flame-retardant properties, the THR and TSP were 35.6% and 7.2% reduced than those of natural wood, a LOI of 34.7%. The phosphorus containing groups in modifier catalyzed dehydration and char formation, while the cured polymer yielded a stable char structure, effectively enhancing flame retardancy.

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Data availability Data is provided within the manuscript.

Declarations

Conflict of interest The authors declare no competing interests.

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